

CHAPTER 9

Ray Optics and Optical Instruments

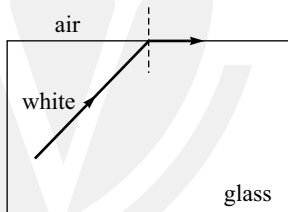
Reflection of Light and Spherical Mirrors

- Given below are two statements:
Statement I: Image formation needs regular reflection and/or refraction.
Statement II: The variety in colour of objects we see around us is due to the constituent colours of the light incident on them.
 In the light of the above statements, choose the most appropriate answer from the options given below: (2024 Re)
 a. Statement I is correct but statement II is incorrect
 b. Statement I is incorrect but Statement II is correct
 c. Both Statement I and Statement II are correct
 d. Both Statement I and Statement II are incorrect
- An object is placed on the principal axis of a concave mirror at a distance of $1.5f$ (f is the focal length). The image will be at, (2020-Covid)
 a. $1.5f$ b. $-1.5f$ c. $3f$ d. $-3f$
- An object is placed at a distance of 40 cm from a concave mirror of focal length 15 cm. If the object is displaced through a distance of 20 cm towards the mirror, the displacement of the image will be: (2018)
 a. 30 cm towards the mirror
 b. 36 cm away from the mirror
 c. 30 cm away the mirror
 d. 36 cm towards the mirror
- Match the corresponding entries of column-1 with column-2. [where m is the magnification produced by the mirror] (2016 - I)

Column-1	Column-2
(A) $m = -2$	(1) Convex mirror
(B) $m = -\frac{1}{2}$	(2) Concave mirror
(C) $m = +2$	(3) Real image
(D) $m = \sqrt{2} / \sqrt{3}$	(4) Virtual image

a. A $\rightarrow$ 2 and 3; B $\rightarrow$ 2 and 3; C $\rightarrow$ 2 and 4; D $\rightarrow$ 1 and 4
 b. A $\rightarrow$ 1 and 3; B $\rightarrow$ 1 and 4; C $\rightarrow$ 1 and 2; D $\rightarrow$ 3 and 4
 c. A $\rightarrow$ 1 and 4; B $\rightarrow$ 2 and 3; C $\rightarrow$ 2 and 4; D $\rightarrow$ 2 and 3
 d. A $\rightarrow$ 3 and 4; B $\rightarrow$ 2 and 4; C $\rightarrow$ 2 and 3; D $\rightarrow$ 1 and 4

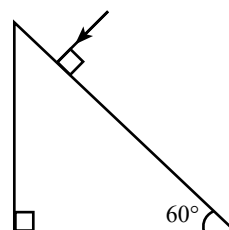
Laws of Refraction; Refractive Index

- Light travels a distance x in time t_1 in air and $10x$ in time t_2 in another denser medium. What is the critical angle for this (2023)
 a. $\sin^{-1}\left(\frac{10t_1}{t_2}\right)$ b. $\sin^{-1}\left(\frac{t_2}{t_1}\right)$
 c. $\sin^{-1}\left(\frac{10t_2}{t_1}\right)$ d. $\sin^{-1}\left(\frac{t_1}{10t_2}\right)$
- ϵ_0 and μ_0 are the electric permittivity and magnetic permeability of free space respectively. If the corresponding quantities of a medium are $2\epsilon_0$ and $1.5\mu_0$ respectively, the refractive index of the medium will nearly be: (2023-Manipur)
 a. $\sqrt{2}$ b. $\sqrt{3}$ c. 3 d. 2
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Which set of colours will come out in air for a situation shown in figure? (2023-Manipur)
 a. Yellow, Orange and Red b. All
 c. Orange, Red and Violet d. Blue, Green and Yellow

Refraction Through Glass Slab and Lateral Shift

- A light ray falls on a glass surface of refractive index $\sqrt{3}$, at an angle 60° . The angle between the refracted and reflected rays would be: (2022)
 a. 120° b. 30° c. 60° d. 90°
- Find the value of the angle of emergence from the prism. Refractive index of the glass is $\sqrt{3}$. (2021)



- a. 30° b. 45° c. 90° d. 60°

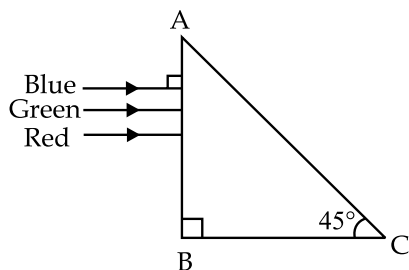
10. A thin prism having refracting angle 10° is made of glass of refractive index 1.42. This prism is combined with another thin prism of glass of refractive index 1.7. This combination produces dispersion without deviation. The refracting angle of second prism should be: (2017-Delhi)
- a. 6° b. 8° c. 10° d. 4°

Real and Apparent Depths

11. An air bubble in a glass slab with refractive index 1.5 (near normal incidence) is 5 cm deep when viewed from one surface and 3 cm deep when viewed from the opposite surface. The thickness (in cm) of the slab is: (2016-II)
- a. 12 b. 16 c. 8 d. 10

Total Internal Reflection and Critical Angle

12. Two transparent media A and B are separated by a plane boundary. The speed of light in those media are 1.5×10^8 m/s and 2.0×10^8 m/s, respectively. The critical angle for a ray of light for these two media is: (2022)
- a. $\tan^{-1}(0.750)$ b. $\sin^{-1}(0.500)$
 d. $\sin^{-1}(0.750)$ d. $\tan^{-1}(0.500)$
13. If the critical angle for total internal reflection from a medium to vacuum is 45° , then velocity of light in the medium is, (2020-Covid)
- a. $\frac{3}{\sqrt{2}} \times 10^8$ m/s b. $\sqrt{2} \times 10^8$ m/s
 c. 3×10^8 m/s d. 1.5×10^8 m/s
14. In total internal reflection when the angle of incidence is equal to the critical angle for the pair of media in contact, what will be angle of refraction? (2019)
- a. 180°
 b. 0°
 c. Equal to angle of incidence
 d. 90°
15. A beam of light consisting of red, green and blue colors is incident on a right angled prism. The refractive index of the material of the prism for the above red, green and blue wavelengths are 1.39, 1.44 and 1.47, respectively



The prism will: (2015 Pre)

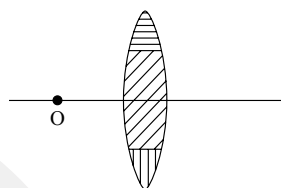
- a. Separate the red color part from the green and blue colors
 b. Separate the blue color part from the red and green colors
 c. Separate all the three colors from one another
 d. Not separate the three colors at all

Refraction at Spherical Surfaces

16. A plano-convex lens of unknown material and unknown focal length is given. With the help of a spherometer we can measure the, (2020-Covid)
- a. Radius of curvature of the curved surface
 b. Aperture of the lens
 c. Refractive index of the material
 d. Focal length of the lens

Lens Formula and Lens Maker's Formula

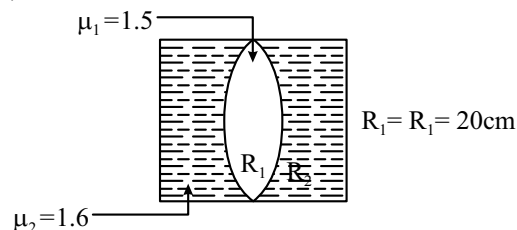
17. A lens is made up of 3 different transparent media as shown in figure. A point object O is placed on its axis beyond $2f$. How many real images will be obtained on the other side? (2023-Manipur)



- a. 2 b. 1
 c. No image will be formed d. 3
18. An object is mounted on a wall. Its image of equal size is to be obtained on a parallel wall with the help of a convex lens placed between these walls. The lens is kept at distance x in front of the second wall. The required focal length of the lens will be: (2023-Manipur)
- a. less than $\frac{x}{4}$ b. more than $\frac{x}{4}$ but less than $\frac{x}{2}$
 c. $\frac{x}{2}$ d. $\frac{x}{4}$
19. A plano-convex lens fits exactly into a planoconcave lens. Their plane surfaces are parallel to each other. If lenses are made of different materials of refractive indices μ_1 and μ_2 and R is the radius of curvature of the curved surface of the lenses, then the focal length of the combination is: (2013)
- a. $\frac{2R}{(\mu_2 - \mu_1)}$ b. $\frac{R}{2(\mu_1 + \mu_2)}$
 c. $\frac{R}{2(\mu_1 - \mu_2)}$ d. $\frac{R}{(\mu_1 - \mu_2)}$

Combination of Lenses

20. In the figure shown here, what is the equivalent focal length of the combination of lenses (Assume that all layers are thin)? (2023)



- a. -50 cm b. 40 cm c. -40 cm d. -100 cm

21. Two thin lenses are of same focal lengths (f), but one is convex and the other one is concave. When they are placed in contact with each other, the equivalent focal length of the combination will be: (2023)

a. Infinite b. Zero c. $\frac{f}{4}$ d. $\frac{f}{2}$

22. A convex lens 'A' of focal length 20 cm and a concave lens 'B' of focal length 5 cm are kept along the same axis with a distance 'd' between them. If a parallel beam of light falling on 'A' leaves 'B' as a parallel beam, then the distance 'd' in cm will be: (2021)

a. 15 b. 50 c. 30 d. 25

23. Two similar thin equi-convex lenses, of focal length f each, are kept coaxially in contact with each other such that the focal length of the combination is F_1 . When the space between the two lenses is filled with glycerine (which has the same refractive index ($\mu = 1.5$) as that of glass) then the equivalent focal length is F_2 . The ratio $F_1 : F_2$ will be: (2019)

a. 2 : 1 b. 1 : 2 c. 2 : 3 d. 3 : 4

24. Two identical glass ($\mu_g = 3/2$) equiconvex lenses of focal length f each are kept in contact. The space between the two lenses is filled with water ($\mu_w = 4/3$). The focal length of the combination is: (2016 - II)

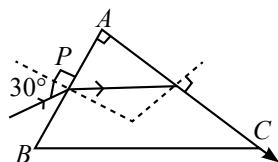
a. $4f/3$ b. $3f/4$ c. $f/3$ d. f

25. Two identical thin plano-convex glass lenses (refractive index 1.5) each having radius of curvature of 20 cm are placed with their convex surfaces in contact at the center. The intervening space is filled with oil of refractive index 1.7. The focal length of the combination is: (2015)

a. -25 cm b. -50 cm c. 50 cm d. -20 cm

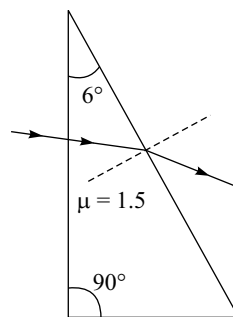
Refraction Through Prism and Dispersion Through Prism

26. A light ray enters through a right angled prism at point P with the angle of incidence 30° as shown in figure. It travels through the prism parallel to its base BC and emerges along the face AC. The refractive index of the prism is: (2024)



a. $\frac{\sqrt{3}}{4}$ b. $\frac{\sqrt{3}}{2}$ c. $\frac{\sqrt{5}}{4}$ d. $\frac{\sqrt{5}}{2}$

27. A horizontal ray of light is incident on the right angled prism with prism angle 6° . If the refractive index of the material of the prism is 1.5, then the angle of emergence will be: (2023-Manipur)



a. 9° b. 10° c. 4° d. 6°

28. A ray is incident at an angle of incidence i on one surface of a small angle prism (with angle of prism A) and emerges normally from the opposite surface. If the refractive index of the material of the prism is m , then the angle of incidence is nearly equal to: (2020)

a. $\frac{2A}{\mu}$ b. μA c. $\frac{2A}{2}$ d. $\frac{A}{2\mu}$

29. The refractive index of the material of a prism is 2 and the angle of the prism is 30° . One of the two refracting surfaces of the prism is made a mirror inwards, by silver coating. A beam of monochromatic light entering the prism from the other face will retrace its path (after reflection from the silvered surface) if its angle of incidence on the prism is: (2018)

a. 30° b. 45° c. 60° d. Zero

30. The angle of incidence for a ray of light at a refracting surface of a prism is 45° . The angle of prism is 60° . If the ray suffers minimum deviation through the prism, the angle of minimum deviation and refractive index of the material of the prism respectively, are: (2016 - I)

a. $45^\circ; \frac{1}{\sqrt{2}}$ b. $30^\circ; \sqrt{2}$ c. $45^\circ; \sqrt{2}$ d. $30^\circ; \frac{1}{\sqrt{2}}$

31. The refracting angle of a prism is A , and refractive index of the material of the prism is $\cot(A/2)$. The angle of minimum deviation is: (2015)

a. $180^\circ - 2A$ b. $90^\circ - A$ c. $180^\circ + 2A$ d. $180^\circ - 3A$

32. The angle of a prism is A . One of its refracting surfaces is silvered. Light rays falling at an angle of incidence $2A$ on the first surface returns back through the same path after suffering reflection at the silvered surface. The refractive index μ , of the prism is: (2014)

a. $2\sin A$ b. $2\cos A$ c. $1/2 \cos A$ d. $\tan A$

Natural Phenomenon

33. During a cloudy day, a primary and a secondary rainbow may be created, then the: (2022 Re)

a. secondary rainbow is due to single reflection and is formed above the primary one.
 b. primary rainbow is due to double internal reflection and is formed above the secondary one.
 c. primary rainbow is due to double internal reflection and is formed below the secondary one.
 d. secondary rainbow is due to double internal reflection and is formed above the primary one.

- a. $\frac{L}{I}$ b. $\frac{L}{I}+1$ c. $\frac{L}{I}-1$ d. $\frac{L+1}{I-1}$

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|------------|
| 1. | (c) | 2. | (d) | 3. | (b) | 4. | (a) | 5. | (a) | 6. | (b) | 7. | (a) | 8. | (d) | 9. | (d) | 10. | (a) |
| 11. | (a) | 12. | (c) | 13. | (a) | 14. | (d) | 15. | (a) | 16. | (a) | 17. | (d) | 18. | (c) | 19. | (d) | 20. | (d) |
| 21. | (a) | 22. | (a) | 23. | (b) | 24. | (b) | 25. | (b) | 26. | (d) | 27. | (a) | 28. | (b) | 29. | (b) | 30. | (b) |
| 31. | (a) | 32. | (b) | 33. | (d) | 34. | (c) | 35. | (d) | 36. | (d) | 37. | (d) | 38. | (d) | 39. | (a) | 40. | (c) |
| 41. | (d) | | | | | | | | | | | | | | | | | | |